

Worked examples: AND, XOR, and the first gradient

Companion to Lecture 1 - numbers you can check by hand

1. Worked AND example

Inputs are (x_1, x_2) in $\{0,1\}^2$. Target $t = x_1 \text{ AND } x_2$. Initialise $w = (0, 0)$, $b = 0$, $\eta = 1$. First example $(1, 1)$, $t = 1$. $z = 0$, prediction 0, a mistake. Update: $w := (1, 1)$, $b := 1$. Second example $(1, 0)$, $t = 0$. $z = 1+0+1 = 2$, prediction 1, a mistake. Update: $w := (0, 1)$, $b := 0$. Continue cycling the four points. A solution you should reach: $w = (1, 1)$, $b = -1.5$ after scaling, or any positive multiple. Check: only $(1,1)$ yields $z \geq 0$.

2. Why XOR breaks the same procedure

Label $t(0,0)=0$, $t(1,1)=0$, $t(0,1)=1$, $t(1,0)=1$. Suppose a line $ax + by + c = 0$ separates them. Then $a*0 + b*0 + c < 0$ so $c < 0$; $a + b + c < 0$; $b + c > 0$; $a + c > 0$. Adding the last two inequalities: $a + b + 2c > 0$. But $a + b + c < 0$ and $c < 0$ imply $a + b + 2c < 0$. Contradiction. No such line exists, so the perceptron updates cycle forever. This is the numerical form of the picture in [xor-decision-regions.png](#).

3. Two-layer XOR by hand

Hidden 1: $w = (1, -1)$, $b = -0.5$ -> fires on $(1,0)$. Hidden 2: $w = (-1, 1)$, $b = -0.5$ -> fires on $(0,1)$. Output: $w = (1, 1)$, $b = -0.5$ -> OR of the two hidden bits. Forward $(1,0)$: $h = (1, 0)$, $y = 1$. Forward $(1,1)$: $h = (0, 0)$, $y = 0$. Forward $(0,0)$: $h = (0, 0)$, $y = 0$. Forward $(0,1)$: $h = (0, 1)$, $y = 1$.

4. Gradient for a linear unit

Let $\hat{y} = w \cdot x + b$ and $L = 0.5 (\hat{y} - t)^2$. Then $dL/dw = (\hat{y} - t) x$ and $dL/db = (\hat{y} - t)$. One SGD step: $w := w - \eta (\hat{y} - t) x$. Compare this to the perceptron rule: both move the weights in the direction of x when the prediction is too small, but SGD uses a continuous error instead of a hard mistake. With a sigmoid, replace $(\hat{y} - t)$ by $(\hat{y} - t) * \hat{y} * (1 - \hat{y})$ if L is squared error, or simply $(\hat{y} - t)$ if L is cross-entropy.

5. Numbers students always mix up

Learning rate η is not a probability. Typical starting values: 0.1 for perceptron on toy data, 0.01 for SGD on a sigmoid MLP, 0.001 for Adam on a deeper net. Epoch means one pass over the training set. Minibatch 32 is a default, not a law. Parameter count for 4-5-4-2: $4*5 + 5 + 5*4 + 4 + 4*2 + 2 = 20 + 5 + 20 + 4 + 8 + 2 = 59$. Bias terms are easy to forget and examiners notice.

6. Glossary for the chat corpus

Perceptron: linear binary classifier with a step activation. Hyperplane: the set $w \cdot x + b = 0$. Linearly separable: some hyperplane classifies every training point correctly. Hidden layer: intermediate units whose outputs are inputs to the next layer. Backpropagation: chain-rule gradients from loss to every weight. ReLU: $\max(0, z)$, the default hidden activation after 2011. Overfitting: train loss low, test loss high. Dropout: randomly zero hidden units at train time. These terms appear in the lecture recording and in the diagram captions, so retrieval should be able to cite all three modalities.